

Weak-star complemented subspaces force Rosenthal sets

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Status

Partial result.

Lefevre–Li–Queffelec–Rodriguez-Piazza conjecture that $\mathcal{C}_\Lambda$ is complemented in L^∞_Λ only in the Rosenthal case $L^\infty_\Lambda = \mathcal{C}_\Lambda$. This packet proves the subcase where the projection is weak-star continuous for the canonical weak-star topology inherited from $L^\infty(G) = L^1(G)^*$.

Source Question

The source paper states in the introduction that Section 5 investigates when $\mathcal{C}_\Lambda$ is complemented in L^∞_Λ , and says:

“We conjecture that this happens only if $\mathcal{C}_\Lambda = L^\infty_\Lambda$, i.e. Λ is a Rosenthal set.”

In this work, we study what are the implications on Λ of the fact that some Banach space X_Λ contains, or not, the space c_0 . In particular, we shall extend (1') and (2').

The paper is organized as follows. In Section 2, we show that if ψ is an Orlicz function which violates the Δ_2 -condition, in a strong sense: $\lim_{x \rightarrow +\infty} \frac{\psi(2x)}{\psi(x)} = +\infty$ (which is the case of Ψ_2), and if X_0 is a linear subspace of L^∞ on which the norms $\|\cdot\|_2$ and $\|\cdot\|_\psi$ are not equivalent, then the closure X of X_0 in L^ψ contains c_0 . It follows that if Λ is not a Sidon set, then $L_\Lambda^{\Psi_2}$ contains c_0 , and *a fortiori* that if $L_\Lambda^{\Psi_2}$ has a finite cotype, then Λ is a Sidon set, which generalizes (1').

In Section 3, we extend (2') by showing that: If Λ is not a set of uniform convergence (*i.e.* if $U_\Lambda \neq \mathcal{C}_\Lambda$), then U_Λ does contain c_0 . In particular, if U_Λ has a finite cotype, then $U_\Lambda = \mathcal{C}_\Lambda$, so $\mathcal{C}_\Lambda$ has a finite cotype and therefore, in view of (2), Λ is a Sidon set. This explains why the proof of (2') in [19] mimicked Bourgain and Milman's.

In Section 4, we use the notion of invariant mean in $L^\infty(G)$. We say that Λ is a *Lust-Piquard set* if, for every function $f \in L_\Lambda^\infty$, the product γf of f with every character $\gamma \in \Gamma$ has a unique invariant mean. Of course, if every $f \in L_\Lambda^\infty$ is continuous (*i.e.* Λ is a Rosenthal set), then Λ is a Lust-Piquard set. F. Lust-Piquard ([27]) showed that there are Lust-Piquard sets which are not Rosenthal sets, and, more precisely, that $\Lambda = \mathbb{P} \cap (5\mathbb{Z} + 2)$, where $\mathbb{P}$ is the set of the prime numbers, is a Lust-Piquard set such that $\mathcal{C}_\Lambda$ contains c_0 (if Λ is a Rosenthal set, $\mathcal{C}_\Lambda$ cannot contain c_0). We construct here another kind of "big" Lust-Piquard set Λ , namely a Hilbert set. Then $\mathcal{C}_\Lambda$ contains c_0 by a result of the second-named author ([22], Theorem 2).

In Section 5, we investigate under which conditions the space $\mathcal{C}_\Lambda$ is complemented in L_Λ^∞ . We conjecture that this happens only if $\mathcal{C}_\Lambda = L_\Lambda^\infty$, *i.e.* Λ is a Rosenthal set. We are only able to show that, under that condition of complementation, $\mathcal{C}_\Lambda$ does not contain c_0 , and, moreover, every $f \in L_\Lambda^\infty$ which is Riemann-integrable is actually in $\mathcal{C}_\Lambda$.

Notation. Throughout this paper, G is a compact abelian group, and $\Gamma = \widehat{G}$ is its (discrete) dual group. The Haar measure of G is denoted by m , and integration with respect to m by dt or dx . We shall write the group structure of Γ additively, so that, for $\gamma \in \Gamma$, the character $(-\gamma) \in \Gamma$ is the function $\overline{\gamma} \in \mathcal{C}(G)$. When G is the circle group $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, we identify, as usual, the character $e_n: t \mapsto e^{int}$ with the integer $n \in \mathbb{Z}$, and so the dual group Γ to $\mathbb{Z}$; the Haar measure is then $dt/2\pi$.

For $f \in L^1(G)$, the Fourier coefficient of f at $\gamma \in \Gamma$ is $\widehat{f}(\gamma) = \int_G f(t) \overline{\gamma(t)} dt$. If X is a linear function subspace of $L^1(G)$, we denote by X_Λ the subspace of those $f \in X$ for which the Fourier coefficients vanish outside of Λ .

When we say that a Banach space X contains a Banach space Y , we mean that X contains a (closed) subspace isomorphic to Y .

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In Section 5 the authors ask whether the existence of a projection from L_Λ^∞ onto $\mathcal{C}_\Lambda$ implies that Λ is a Rosenthal set, and note that they cannot answer this even for norm-one projections.

Define

$$g(x) = \frac{1}{2} \left(\limsup_{n \rightarrow \infty} f_n(x) + \liminf_{n \rightarrow \infty} f_n(x) \right), \quad x \in G.$$

It is clear that $g(x) = g(x+h)$, for all $x \in G$ and for all $h \in H$. Since $V_n \subseteq W_n$, we have from (4) that $|f_n(x) - f(x)| \leq 1/n$, for all $x \in K_n$. If $x \in B = \bigcup_n K_n$, then there exists N such that $x \in K_n$, for all $n \geq N$. Therefore $|f_n(x) - f(x)| \leq 1/n$, for all $n \geq N$, and $g(x) = f(x)$. So $f = g$ almost everywhere.

In order to finish the proof we are going to see that every point of B is a point of continuity of g , and so g is Riemann-integrable. Let x be in B . Given $\varepsilon > 0$, there exists N such that $1/N \leq \varepsilon$ and $x \in K_n$, for all $n \geq N$. We are going to prove

$$|g(x) - g(x+y)| \leq \varepsilon, \quad \text{for every } y \in V_N. \quad (5)$$

So g will be continuous at x .

Take $n \geq N$, and $y \in V_N$. For every $z \in V_n$ we have $x+y+z \in W_N$, and $|f(x) - f(x+y+z)| \leq 1/N$. By the definition of f_n we get $|f(x) - f_n(x+y)| \leq 1/N$, for every $n \geq N$. Then we obtain (5) easily, since $f(x) = g(x)$. $\square$

Remarks. 1) Actually the proof shows that if Λ is a Lust-Piquard set and if there exists a surjective projection $Q: L_\Lambda^\infty \rightarrow \mathcal{C}_\Lambda$ which commutes with translations, then Λ is a Rosenthal set.

2) Talagrand's work [45] uses Martin's axiom, and, in [46], another axiom, called L . But these axioms do not intervene in the results we use (they are needed to obtain Riemann-integrability from the weak measurability of translations: see [46], Theorem 15-4).

3) F. Lust-Piquard and W. Schachermayer ([29], Corollary IV.4 and Proposition IV.15; see also [11], Theorem V.1, Corollary VI.18, and Example VIII.10) showed that if $L^1(G)/L_{\Gamma \setminus (-\Lambda)}^1$ does not contain ℓ_1 (which is equivalent to L_Λ^∞ having the weak Radon-Nikodym property [46], Corollary (7-3-8)), then $L_\Lambda^\infty = RI_\Lambda$. Hence Λ must be a Rosenthal set if L_Λ^∞ has the weak Radon-Nikodym property and there exists a projection from L_Λ^∞ onto $\mathcal{C}_\Lambda$. However, a direct proof is available. For a more general result, see [11], Example following Proposition VII.6.

4) The first part of the proof is the same as the one used by A. Pelczyński ([34], Cor. 9.4 (a)) to show that $A(\mathbb{D}) = \mathcal{C}_\mathbb{N}$ is not complemented in $H^\infty = L_\mathbb{N}^\infty$.

Question. When Λ is not a Rosenthal set, or, merely when $\mathcal{C}_\Lambda$ contains c_0 , how big can $L_\Lambda^\infty/\mathcal{C}_\Lambda$ be?

References

- [1] J. Bourgain, Quelques propriétés linéaires topologiques de l'espace des séries de Fourier uniformément convergentes, Séminaire d'Initiation à

Definitions

Let G be a compact abelian group and let $\Gamma = \widehat{G}$ be its discrete dual group. For $\Lambda \subseteq \Gamma$, write

$$L_\Lambda^\infty = \{f \in L^\infty(G) : \widehat{f}(\gamma) = 0 \text{ for } \gamma \notin \Lambda\}$$

and

$$\mathcal{C}_\Lambda = \mathcal{C}(G) \cap L_\Lambda^\infty.$$

The set Λ is a Rosenthal set when $L_\Lambda^\infty = \mathcal{C}_\Lambda$.

Idea of the Proof

The only extra hypothesis is weak-star continuity. The continuous trigonometric polynomials with spectrum in Λ are weak-star dense in L_Λ^∞ . A weak-star continuous projection onto $\mathcal{C}_\Lambda$ fixes $\mathcal{C}_\Lambda$, hence fixes those polynomials, and therefore fixes their weak-star closure. Thus it must be the identity on all of L_Λ^∞ . Since its range is $\mathcal{C}_\Lambda$, the two spaces are equal.

Theorem

Theorem 1. *Let G be a compact abelian group, $\Gamma = \widehat{G}$, and $\Lambda \subseteq \Gamma$. Suppose there is a bounded projection*

$$P : L_\Lambda^\infty \longrightarrow \mathcal{C}_\Lambda$$

which is continuous for the weak-star topology induced by the predual $L^1(G)$ of $L^\infty(G)$. Then $L_\Lambda^\infty = \mathcal{C}_\Lambda$. Equivalently, Λ is a Rosenthal set.

Proof. Let $E_\Lambda = \text{span}\{\gamma : \gamma \in \Lambda\}$, the trigonometric polynomials with spectrum in Λ . We first recall the standard duality fact that the weak-star closure of E_Λ in $L^\infty(G)$ is exactly L_Λ^∞ .

Indeed, the annihilator of E_Λ in $L^1(G)$ is

$$E_\Lambda^\perp = \{h \in L^1(G) : \int_G \gamma h \, dm = 0 \text{ for every } \gamma \in \Lambda\}.$$

Taking the annihilator again in $L^\infty(G)$ gives precisely the functions whose Fourier coefficients vanish outside Λ , i.e. $(E_\Lambda^\perp)^\perp = L_\Lambda^\infty$. By the bipolar/annihilator description of weak-star closure, E_Λ is weak-star dense in L_Λ^∞ .

Now let $f \in L_\Lambda^\infty$. Choose a net (p_i) in E_Λ which converges to f in the weak-star topology. Since $E_\Lambda \subseteq \mathcal{C}_\Lambda$ and P is a projection onto $\mathcal{C}_\Lambda$, we have $Pp_i = p_i$ for every i . Weak-star continuity of P gives

$$Pf = w^* - \lim_i Pp_i = w^* - \lim_i p_i = f.$$

But $Pf \in \mathcal{C}_\Lambda$, so $f \in \mathcal{C}_\Lambda$. Since f was arbitrary, $L_\Lambda^\infty \subseteq \mathcal{C}_\Lambda$, while the reverse inclusion is automatic. Hence $L_\Lambda^\infty = \mathcal{C}_\Lambda$. $\square$

Verification and Limitations

The proof uses only the canonical duality $L^\infty(G) = L^1(G)^*$ and the definition of spectral subspaces. It works for arbitrary compact abelian groups, with no metrizability assumption, because weak-star density is stated using nets.

This is not a full solution of the source conjecture. The source asks about arbitrary bounded projections, and explicitly mentions that even norm-one projections are open. A bounded projection onto a norm-closed subspace of a dual space need not be weak-star continuous, so the theorem handles only the normal/weak-star-continuous subcase.

Novelty Check

Local index searches for arXiv:0912.3133, $\mathcal{C}_\Lambda$, L_Λ^∞ , Rosenthal sets, complemented subspaces, and projection terms found no prior packet. Web searches on 2026-06-16 for the exact projection/Rosenthal phrases found the source paper but no later exact answer. The result is a direct subcase proof rather than a literature identification.

References

References

- [1] P. Lefevre, D. Li, H. Queffelec, and L. Rodriguez-Piazza, *Some translation-invariant Banach function spaces which contain c_0* , arXiv:0912.3133.